

COA + DSD Integrated Simulator

Short Project Overview • Semester 3 Computer Engineering

1. Project Overview

A software-only interactive learning and simulation platform that combines Digital System Design (DSD) and Computer Organization & Architecture (COA). The project demonstrates the complete path from basic logic gates and digital circuits to ALU, memory, CPU concepts and 8086 program execution.

2. Problem Statement

DSD and COA practicals are often performed as separate experiments, making it difficult for students to visualize how digital circuits become components of a computer system. The proposed application provides one interactive environment for learning, simulation, experimentation and visualization.

3. Main Objectives

- Create an interactive DSD and COA simulation platform.
- Visualize theoretical concepts instead of only displaying final answers.
- Connect DSD building blocks with COA components such as ALU, registers and CPU.
- Provide step-by-step visualization for important algorithms and processor operations.
- Create a reusable virtual laboratory for practical experiments and revision.

4. Major Modules

Module	Main Features
DSD — Basics	Number systems, binary arithmetic, Gray/BCD/ASCII/Excess-3 conversion
Logic & Boolean	Logic gates, truth tables, Boolean algebra, SOP/POS, De Morgan, universal gates
K-Map	2–4 variable minimization and simplified expressions
Combinational	Adders, subtractors, MUX/DEMUX, encoder/decoder, 2-bit comparator, BCD adder
Sequential	SR/JK/D/T flip-flops, conversions, counters and shift registers
COA Algorithms	Booth, restoring division, non-restoring division and IEEE 754
ALU	Arithmetic, logic, shifting and comparison operations
Memory	RAM, memory hierarchy, cache mapping and basic paging/address translation
8086	Registers, instructions, arithmetic, code conversion, arrays, GCD/LCM, factorial and data transfer
Advanced COA	Instruction cycle, control unit, pipeline and basic multicore/Flynn visualization
Learning Tools	Experiment mode, step-by-step results, quiz and saved work

5. DSD + COA Integration

The main strength of the project is the relationship between the two subjects: logic gates form combinational circuits; adders, multiplexers and comparators contribute to an ALU; flip-flops form storage elements and registers; counters support sequencing; memory and control logic build the larger computer architecture.

Core demonstration: Logic Gates → Combinational Circuits → Sequential Circuits → ALU → Registers → Memory → Control Unit → CPU → 8086 execution.

6. Recommended Technology Stack

Technology	Where it is used
Java 21	All simulation logic, algorithms, CPU/memory models and application classes
JavaFX	GUI, circuit editor, tables, diagrams, LEDs, animations and 8086 interface

Technology	Where it is used
Maven	Dependency management, build and packaging
SQLite	Saved circuits, programs, experiment history, quiz scores and user data
Git + GitHub	Team collaboration, version control and backup
draw.io	Architecture, UML, flowcharts and documentation diagrams
Logisim Evolution	Reference and verification during circuit development; not required to run final app
Figma (optional)	UI planning and mockups

7. Suggested Application Structure

Dashboard → DSD Lab → COA Lab → Circuit Builder → Algorithm Visualizer → ALU → Memory/Cache → 8086 Emulator → Experiments → Quiz.

The circuit builder should be a simplified Logisim-style environment. Java performs the logic calculations while JavaFX renders gates, wires, inputs, outputs and animations.

8. Key Advantages / Pros

Integrated learning: Combines two closely related subjects in one platform.

Highly visual: Students can see circuits, data flow, algorithm steps and processor operations.

Practical-oriented: Covers many DSD and COA laboratory experiments in a virtual environment.

Software-only: No ICs, breadboards, sensors or special hardware are required.

Safe and repeatable: Experiments can be repeated without damaging components or requiring physical setup.

Low-cost: Uses common development software and open/reference tools.

Step-by-step learning: Algorithms such as Booth and division can show intermediate states.

Strong DSD–COA connection: Shows how gates and sequential circuits contribute to ALU, registers and CPU.

Expandable: Future modules such as pipelining, multicore, VLIW and more instructions can be added.

Good demonstration value: Interactive UI, animations and a circuit builder make the project easy to demonstrate.

Team-friendly: DSD, COA, emulator, UI and database work can be divided among team members.

Reusable: The finished software can be used for practical preparation, revision and demonstrations.

9. Academic Advantages

- Directly maps to the DSD and COA syllabus and practical lists.
- Provides a strong basis for explaining algorithms during viva.
- Demonstrates programming, simulation, UI design and system-level thinking.
- Shows practical understanding rather than only theoretical knowledge.
- Creates a clear connection between digital logic and computer architecture.

10. Development Difficulty

Overall difficulty: High for Semester 3, but achievable with a team. The individual algorithms are manageable; the most difficult parts are the drag-and-drop circuit simulator, sequential-state simulation and 8086 emulator. A polished project should prioritize fewer features that work reliably instead of implementing every possible topic superficially.

11. Suggested Team Division

Member	Primary responsibility
1	DSD basics, Boolean algebra, K-map and combinational circuits
2	Flip-flops, counters, shift registers and circuit simulation
3	COA algorithms, ALU, memory and cache
4	8086 emulator, JavaFX UI, integration and database

12. Limitations

- A full Logisim-level circuit editor is complex and should be simplified for a semester project.
- A complete production-grade 8086 emulator is beyond the required scope; support a focused instruction set.
- Simulation accuracy and testing are important, especially for sequential circuits and CPU state changes.
- The project should be scoped according to the team's available development time.

13. Future Scope

- More 8086 instructions and debugging features.
- Advanced pipeline and hazard simulation.
- Multicore and superscalar visualization.

- PLA/PAL and memory-device simulation.
- More advanced cache and virtual-memory scenarios.
- Cloud/web version for browser-based access.
- Automatic experiment reports and performance analytics.

14. Overall Evaluation

Parameter	Rating
Technical depth	9.5 / 10
DSD + COA integration	10 / 10
Syllabus relevance	9.5 / 10
Innovation	9 / 10
Visual/demo potential	10 / 10
Practical usefulness	10 / 10
Semester 3 feasibility	8 / 10
Overall project potential	9 / 10

15. Final Project Pitch

“An integrated software-based virtual laboratory that demonstrates the relationship between Digital System Design and Computer Organization & Architecture, allowing students to build and simulate digital circuits, execute COA algorithms, visualize ALU and memory operations, and run basic 8086 programs in one interactive environment.”